

CHARLEVILLE, Australia

WMO: 945100

Lat: 26.4139S

Lon: 146.2558E

Elev: 303

StdP: 97.73

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.5	2.0	-11.8	1.4	22.8	-9.2	1.8	20.0	8.4	19.9	7.6	19.0	0.6	170	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.2	39.5	19.3	38.1	19.1	36.7	19.0	23.7	29.2	23.1	29.0	22.5	28.7	4.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.6	18.0	25.2	21.7	17.0	24.9	20.8	16.1	24.8	73.0	29.0	70.2	28.6	67.8	28.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.5	6.8	-2.0	41.8	1.3	1.7	-3.0	43.0	-3.7	44.0	-4.5	45.0	-5.4	46.2		
(5)			-3.1	26.9	1.6	3.6	-4.3	29.5	-5.3	31.6	-6.2	33.6	-7.4	36.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	28.9	27.6	25.7	21.3	16.3	13.1	12.4	14.3	18.9	22.7	25.6	28.0		
	DBStd	6.75	3.01	3.05	3.05	3.23	3.39	3.41	3.17	3.57	3.97	3.73	3.53	3.33		
	HDD10.0	24	0	0	0	0	1	9	11	4	0	0	0	0		
	HDD18.3	620	0	0	1	11	80	160	186	134	41	7	1	0		
	CDD10.0	4111	586	492	488	338	197	101	84	138	268	395	467	557		
	CDD18.3	1664	327	258	229	99	18	2	2	9	58	144	218	299		
	CDH23.3	21034	4389	3234	2659	1076	204	23	27	168	829	1800	2678	3948		
(13)	CDH26.7	10896	2548	1750	1319	369	26	1	1	38	317	851	1416	2262		
(14)	Wind	WSAvg	3.3	3.8	3.8	3.4	3.0	2.6	2.6	2.9	3.2	3.6	3.8	3.9		
(15) Precipitation	PrecAvg	474	73	64	48	26	32	24	23	20	25	29	49	57		
	PrecMax	917	267	295	284	248	199	137	99	92	128	124	302	196		
	PrecMin	245	3	0	0	0	0	0	0	0	0	0	1	2		
	PrecStd	160	60	64	54	40	40	27	26	21	31	26	55	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	40.9	38.2	33.8	29.2	25.3	26.0	30.5	35.1	37.2	40.0	41.1		
		MCWB	20.2	20.5	19.5	18.7	16.5	15.0	13.8	15.0	15.8	17.1	18.4	18.9		
	2%	DB	39.1	38.2	36.1	31.7	27.1	23.8	23.8	27.1	32.1	35.0	37.6	39.1		
		MCWB	19.7	19.7	19.3	17.8	15.7	14.4	13.0	13.9	15.2	16.8	18.0	18.7		
	5%	DB	37.7	36.3	34.5	30.3	25.6	22.2	22.1	25.0	30.2	33.3	35.6	37.2		
		MCWB	19.8	19.7	18.9	16.9	15.0	13.8	12.5	12.9	14.6	16.5	17.9	18.5		
	10%	DB	36.1	34.6	32.9	28.9	24.1	20.4	20.3	23.0	28.2	31.5	33.7	35.7		
		MCWB	19.7	19.6	18.8	16.3	14.2	12.8	11.8	12.1	14.4	16.0	17.4	18.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.9	24.7	23.6	21.9	19.6	17.3	18.2	17.5	19.6	21.2	23.2	23.7		
		MCDB	30.4	29.7	27.6	27.9	23.3	21.1	22.2	24.9	24.4	28.2	27.0	30.3		
	2%	WB	23.7	23.7	22.6	20.4	18.0	16.2	15.8	15.7	17.9	19.8	21.2	22.8		
		MCDB	29.5	29.1	27.7	26.8	22.9	20.3	19.1	22.8	24.4	27.4	27.3	29.0		
	5%	WB	23.0	23.0	21.8	19.3	16.9	15.3	14.3	14.5	16.9	18.9	20.5	22.1		
		MCDB	29.4	28.8	27.7	26.2	22.5	19.7	19.1	21.6	25.0	26.9	27.6	28.3		
	10%	WB	22.3	22.3	21.0	18.2	15.8	14.2	12.9	13.3	15.8	18.0	19.7	21.3		
		MCDB	29.4	28.7	27.8	25.1	22.0	18.7	19.0	21.1	24.9	26.5	27.8	28.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.0	12.7	13.9	14.5	13.5	15.0	16.2	15.9	14.9	13.4	12.8		
		MCDBR	14.0	14.4	14.4	15.1	15.2	14.7	16.6	17.8	17.8	16.7	15.7	14.8		
		MCWBR	4.4	4.5	4.9	5.6	6.4	6.9	8.0	8.0	7.2	6.2	5.4	5.1		
	5% WB	MCDBR	10.6	10.5	10.4	11.3	11.4	10.6	11.9	13.6	13.4	12.9	11.3	11.2		
		MCWBR	3.7	3.5	4.0	4.6	5.3	5.5	6.4	6.7	6.3	5.9	4.6	4.1		
(40) Clear-Sky Solar Irradiance	taub		0.360	0.350	0.331	0.313	0.296	0.288	0.283	0.297	0.334	0.347	0.369	0.364		
	taud		2.543	2.582	2.627	2.633	2.621	2.646	2.636	2.567	2.458	2.458	2.448	2.503		
	Ebn at Noon		982	976	967	941	915	903	922	943	947	968	967	981		
	Edh at Noon		111	104	94	85	78	73	76	89	109	116	121	115		
(44) All-Sky Solar Radiation	RadAvg		7.40	6.93	6.17	5.21	4.17	3.53	3.98	4.96	6.04	6.96	7.27	7.44		
	RadStd		0.40	0.55	0.35	0.33	0.20	0.25	0.28	0.27	0.43	0.52	0.63	0.54		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.66	N/A	N/A	+0.88	N/A	N/A	N/A	N/A	+166
(47) Regional (1 neighbor)		+0.66	N/A	N/A	+0.88	N/A	N/A	N/A	N/A	+166

Nomenclature: See separate page